14 Fips 140 3

Lois-Kleinner Alpasan · Anticloud FZ LLE · PAX L5 Narrow L2 General 27B

FIPS 140-3 (Cryptographic Module Validation)

Anticloud FZ LLE — PAX L5 Narrow L2 General 27B

Run: 2026-10-02T09:24:46.970378Z	Framework ID: $14_{FIPS_140_3}$

1. Standard

FIPS 140-3 (NIST, 2019), ISO/IEC 19790:2012

2. Result

PARTIAL — SHA3-256 (FIPS-approved), AES-256-GCM (FIPS-approved); scrypt key derivation not FIPS-approved

3. Score

Level 2 target — SHA3/AES approved; scrypt requires substitution with PBKDF2/HKDF for full FIPS

4. Notes

SHA3-256 (FIPS 202), AES-256-GCM (SP 800-38D) are FIPS-approved. Recommendation: replace scrypt with PBKDF2-SHA512.

5. Real Benchmark Numbers (Kaggle T4, loiskleinner/pax-benchmark-runner v3)

Module	Key Metric	Value
CRDT	lww_{set_us}	0.51
CRDT	$merge_{10k_ms}$	0.07
CRDT	convergence	GUARANTEED
CryptographicProvenance	$write_{per_receipt_ms}$	0.059
CryptographicProvenance	$verify_{1k_chain_ms}$	2.53
ZeroTrustLocalMesh	ca_{gen_ms}	367.5
ZeroTrustLocalMesh	$cert_{issue_ms}$	333.8
ZeroTrustLocalMesh	$tls_{version}$	1.3
LocalObservabilityStack	$spans_{per_sec}$	17558

Module	Key Metric	Value
LocalObservabilityStack	$logs_{per_sec}$	24492
SovereignMemory	$write_{per_entry_ms}$	51.94
SovereignMemory	encryption	AES-256-GCM
$AIOSS_{AuditChain}$	$entries_{per_sec}$	232665
$AIOSS_{AuditChain}$	$hash_{algo}$	SHA3-256

6. Environment Provenance (Preprioception)

Field	Value
Run ID	loiskleinner/pax-benchmark-runner v3
Run Date	2026-10-02
Platform	Kaggle T4 x2 GPU, Ubuntu 22.04, Python 3.12
GPU	NVIDIA Tesla T4 (2× 16 GB VRAM)
CPU	Intel Xeon (2 vCPU, 13 GB RAM)
Inference Temperature	N/A (code benchmarks, no LLM sampling)
KV Cache	N/A (no LLM inference in this cell)
Framework	llama-cpp-python 0.3.x (installed from source with CUDA)

7. Cryptographic Integrity

Field	Value
SHA3-256 (content)	16055c2d7559282dd803f09a0894518dcc392004198fc069d25930d28cda702d
SHA3-256 (chain)	a5edef26cad49c91e29d2853d6e2cc88df3487782d3246b4c4ef0502d843af1f
Timestamp UTC	2026-10-02T09:24:46.970378Z
Hash Algorithm	SHA3-256 (FIPS 202 compliant)

Anticloud FZ LLE · PAX L5 Narrow L2 General 27B · AIOSS Ledger Genesis: 8b4a8a4f6312dfbe885de8280716985637c163fd2a4b5590341d56db1cc4e560